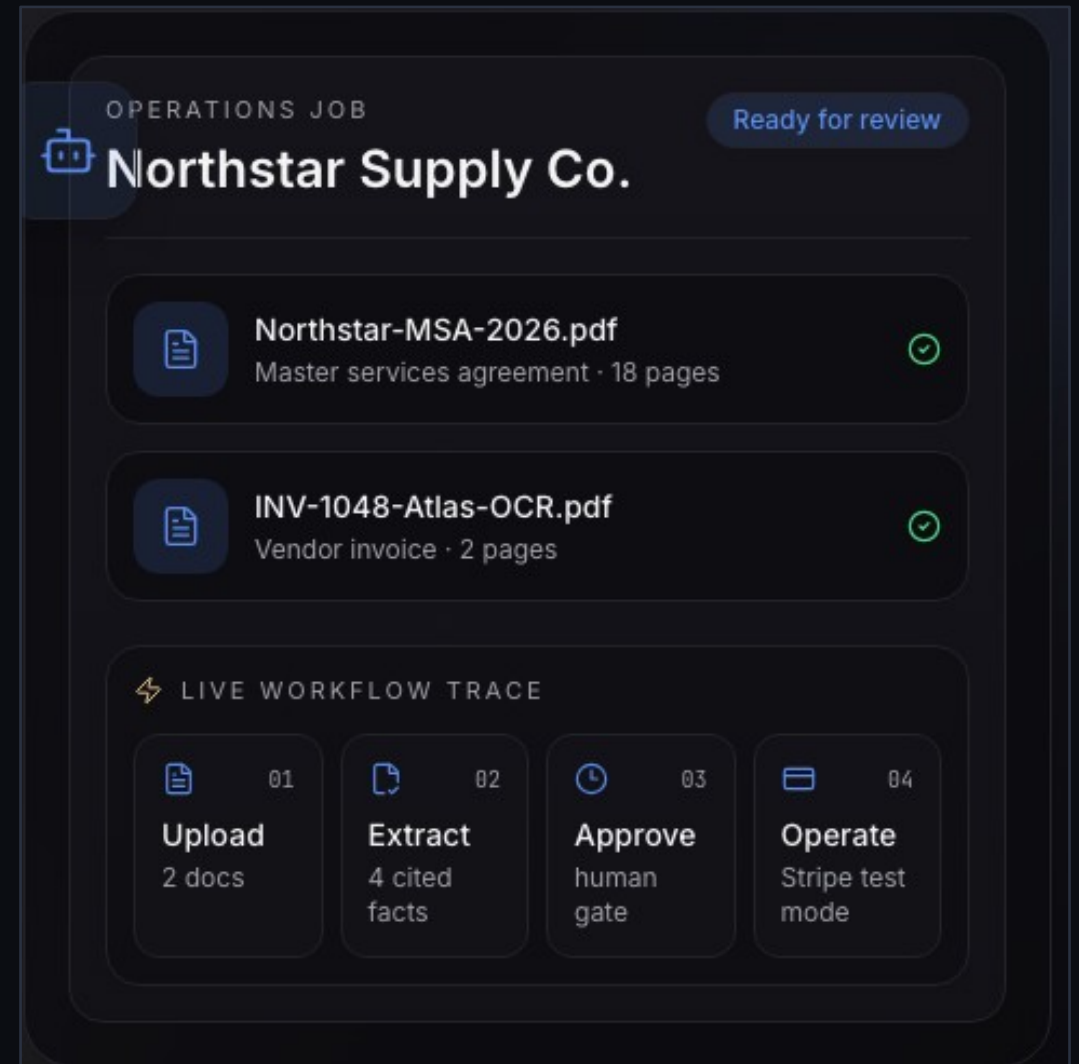


Paperline

Cited document intelligence with human-controlled operations

Full-stack Applied AI portfolio case study ·
Harrison Olvera

Synthetic recruiter demo · Stripe test mode



Next.js · TypeScript · Clerk · Supabase/pgvector · OpenAI · Stripe

Document AI needs evidence, structure, and control—not just plausible answers.

Evidence

Can the reviewer trace a field or answer back to the source page?

Paperline's answer

Quoted snippets + page-level citations

Structure

Can the output enter a repeatable workflow instead of remaining chat text?

Paperline's answer

Schema-guided, runtime-validated extraction

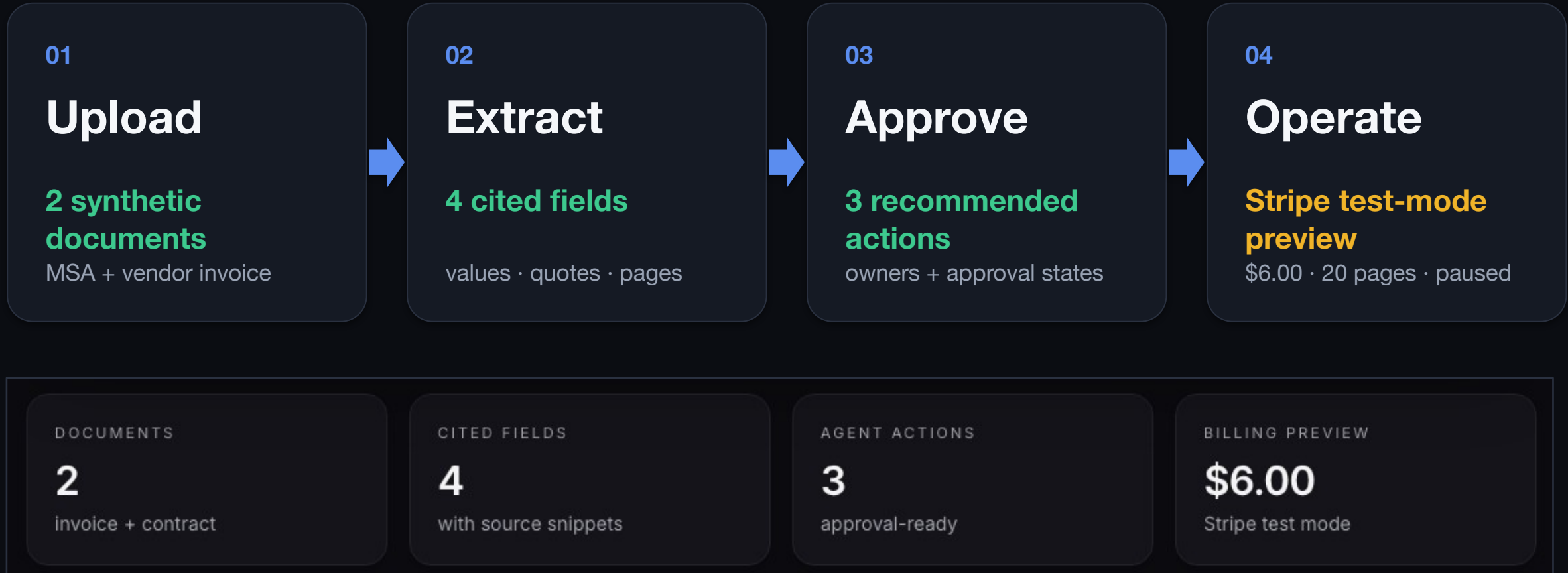
Control

Can the system prepare actions without silently spending or changing external systems?

Paperline's answer

Human approval before spend or provisioning

From upload to approval-ready operations



Live demo route: paperline-xi.vercel.app/ops-agent — one reviewer-friendly surface for the whole product story.

Facts with receipts

CITED EXTRACTION

Facts with receipts

INVOICE TOTAL98%

\$2,850.00

"Amount due: \$2,850.00. Payment due by February 15, 2026."

INV-1048-Atlas-OCR.pdf · p.1

PAYMENT DUE DATE96%

Feb 15, 2026

"Net 30 payment terms apply to this invoice."

INV-1048-Atlas-OCR.pdf · p.1

CONTRACT TERM93%

12 months with auto-renewal

"The initial term shall be twelve months and renew automatically unless cancelled 30 days before renewal."

Northstar-MSA-2026.pdf · p.12

HUMAN REVIEW FLAG91%

Auto-renewal clause

"Cancellation notice must be delivered no later than thirty days prior to the renewal date."

Northstar-MSA-2026.pdf · p.12

HUMAN APPROVAL BOUNDARY

Recommended actions

1 Approve invoice for payment

Finance lead

Approval required

Invoice is payable after human approval. Hermes will not initiate spend until approval is recorded.

2 Create renewal reminder

Operations

Ready to schedule

Set a reminder 45 days before renewal so the team has time to cancel or renegotiate.

3 Ask vendor to clarify auto-renewal

Hermes draft

Draft ready

Prepare a vendor email asking whether the clause can be changed to opt-in renewal.

- Page-level references**

Every field cites its source document and page.
- Quoted source snippets**

The exact supporting text ships with the value.
- Confidence = review signal**

Labels prioritize review — not a guarantee.
- Human review flag**

Auto-renewal clause is explicitly flagged for a person.

Displayed documents, values, and confidence labels are synthetic fixtures

The model extracts; the application controls.

MODEL-ASSISTED

- Document classification
- Schema-guided extraction
- OCR-assisted processing of scans and images
- Grounded answer generation with citations

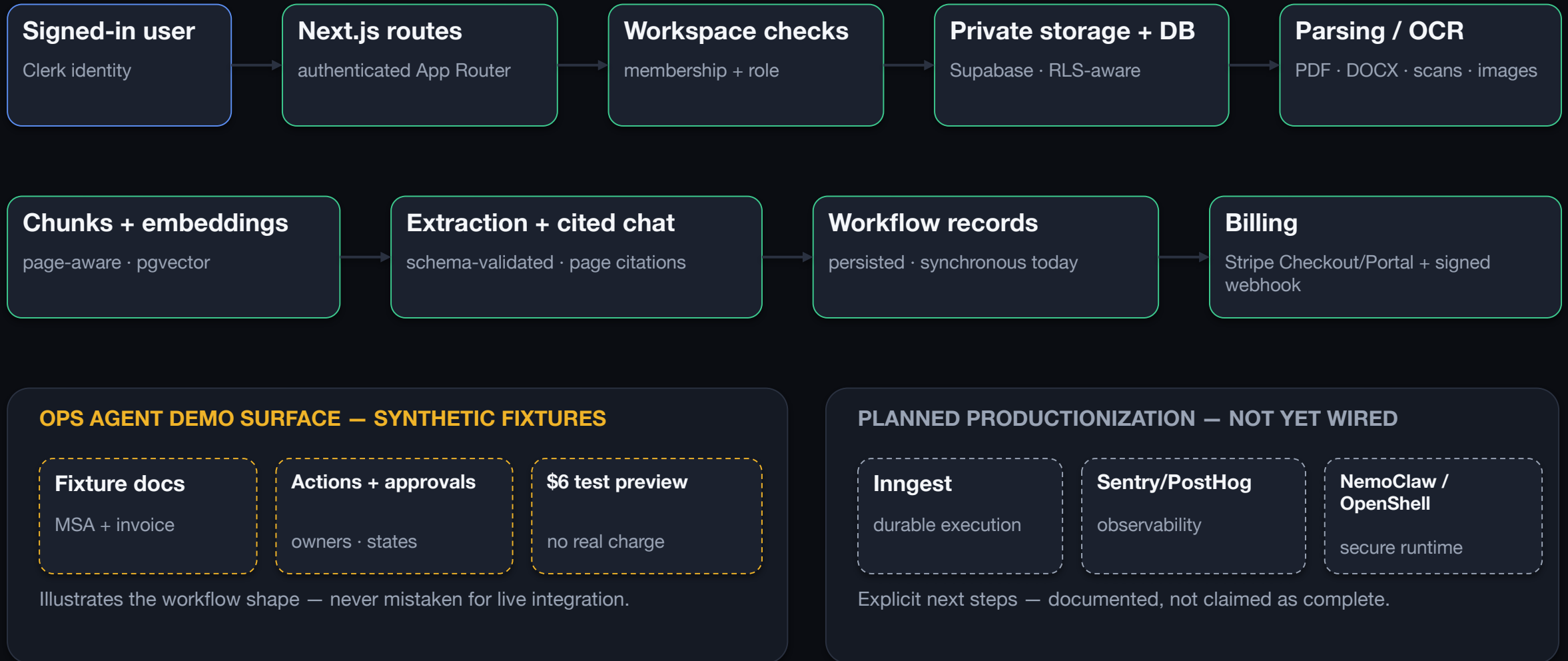
Probabilistic output is treated as a proposal:
inspectable, cited, and validated before use.

DETERMINISTIC APPLICATION CONTROLS

- Authentication + workspace authorization
- Runtime schema validation of extraction output
- Private storage and persisted records
- Usage and billing rules
- Approval ownership and state

Model judgment never owns identity, money, tenancy, or state.

Full-stack document intelligence architecture



Service-role DB access bypasses RLS — route-level workspace filters remain an explicit, tested control.

A deterministic extraction regression harness

Sample-prediction baseline used to validate the scorer — not live-model or production accuracy

3

synthetic document
cases

20

labeled fields

40.00%

exact accuracy

75.00%

normalized accuracy

94.12%

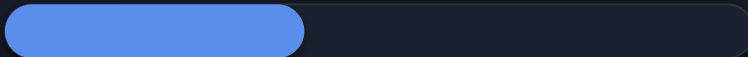
presence F1

85.71%

list-item F1

WHY EXACT MATCHING UNDERSTATES QUALITY

Exact



40.00% (8/20)

Normalized



75.00% (15/20)

Same predictions, scored two ways — normalization isolates formatting noise.

TYPE-AWARE NORMALIZATION

Casing and whitespace
Currency and numbers
Dates and booleans
Order-insensitive lists

Agentic value without uncontrolled action

STRIPE

Test-mode business operation

Mode

Stripe test mode

Customer

Northstar Supply Co.

Usage

20 pages processed

Preview

\$6.00 usage preview

Waiting for human approval: the agent prepares the billing/provisioning step, but does not spend without approval.

HERMES ACTION LOG

Operator trace

1

Paperline received two documents and grouped them into one operations job.

2

Extraction templates identified invoice fields, contract dates, and clause risks.

3

Hermes generated an approval checklist and vendor follow-up draft.

4

Stripe test-mode usage preview calculated 20 processed pages for the workspace.

5

External spend/provisioning is paused until a human approves the action.

NVIDIA

Secure runtime path

✓

Hermes operates from cited Paperline extraction output instead of unbounded document access.

✓

NVIDIA NemoClaw/OpenShell is the enterprise runtime path for sandboxing, credential brokering, and network policy.

✓

Every external action stays auditable and behind an approval boundary before spend or outbound changes.

DEMONSTRATED ON THE SITE

- Visible action owners and states
- Stripe test-mode usage preview
- External spend/provisioning paused

IMPLEMENTED IN THE REPOSITORY

- Stripe Checkout / Portal routes
- Signature-verified subscription webhook
- Persisted workflow records

VALIDATED PATH — NOT YET INTEGRATED

- NVIDIA NemoClaw/OpenShell policy-governed Hermes sandbox
- Controlled network access
- Managed credential/inference handling

NVIDIA NemoClaw/OpenShell is a validated secure-runtime direction, not a completed Paperline integration. Source:

NVIDIA/NemoClaw and NVIDIA/OpenShell official GitHub documentation, verified 2026-07-14.

What I would productionize next

1 Durable execution

Move synchronous workflow execution to Inngest with idempotency, retries, resume, and dead-letter handling.

2 Persisted approvals

Add explicit approval policies and state transitions from proposed through completed/failed.

3 Evaluation expansion

Capture live predictions against a larger versioned, privacy-safe benchmark; add retrieval/citation evaluation.

4 Observability + isolation

Design safe Sentry/PostHog events and validate the NemoClaw/OpenShell path without logging document content.

What Paperline demonstrates

- ✓ Full-stack AI product engineering
- ✓ Cited retrieval and schema-guided extraction
- ✓ Multi-tenant auth/data-boundary thinking
- ✓ Human-in-the-loop agent design
- ✓ Evaluation, billing, and productionization judgment

Harrison Olvera · Applied AI / Agentic AI / LLM Engineering

All demo data synthetic · Stripe test mode · no customer, revenue, or compliance claims

SEE IT YOURSELF

Live Ops Agent

paperline-xi.vercel.app/ops-agent

GitHub

github.com/HarrisonBlake01/paperline

Claims, sources, and boundaries

Implemented repository paths	src/lib/pipeline · src/lib/parsing · src/lib/ai/{ocr,extract,chat} · src/lib/auth/workspace.ts · src/app/api/workflows · src/app/api/billing/{checkout,portal} · src/app/api/webhooks/stripe · supabase/migrations/0001_init.sql, 0002_rls.sql
Synthetic demo boundary	Ops Agent route uses static fixtures from src/lib/ops-agent-demo.ts: Northstar Supply Co., document names, values, confidence labels, action states, operator trace, and the \$6.00 / 20-page Stripe test-mode preview. No live extraction, payment, or external action.
Evaluation caveat	Metrics are a deliberately imperfect sample-prediction baseline validating the deterministic scorer (3 cases, 20 fields). Not production, live-model, or customer accuracy.
NVIDIA sources	github.com/NVIDIA/NemoClaw · github.com/NVIDIA/OpenShell · docs.nvidia.com/nemoclaw · docs.nvidia.com/openshell — NemoClaw lists Hermes as a supported agent. Verified 2026-07-14.
Compliance	Paperline does not claim HIPAA, SOC 2, or legal certification. No customer, revenue, or business-outcome claims.